

Domain-Scoped AFL Chat & Prediction Assistant

Executive Report & Monitoring Plan | Week 3 Day 5 Capstone

Qasim, BSSE 2022 (2022-SE-49), UET Lahore | Due 18 Sept 2026

1. Product Goal

The assistant answers AFL fan questions in two modes: **retrieval** (exact stats, records, and head-to-head history looked up from 2012-2018 match and player data) and **prediction** (a trained model's probability estimate for who wins an upcoming match). A LangGraph router keeps the two modes structurally separate so a factual question can never be answered with a guess, and a prediction can never be presented without its probability and disclaimer. The assistant refuses anything outside AFL (other sports, general trivia, prompt-injection attempts) with a redirect back into scope rather than a bare no, and answers questions about AFL history and rules that need no lookup at all.

2. Architecture

Router → {**Retrieval** | **Prediction** | **Factual** | **Refusal**} → **Validation** → {**Clarify** | **Fallback** | **Format**} → **Response**. An LLM (Gemini 3.5 Flash-Lite live, or a deterministic offline stand-in for reproducible grading) classifies intent once, at the router; every other node is plain Python calling the Day 2 scikit-learn prediction models or the Day 3 pandas lookup tools directly — no second model call, so grounding and disclaimer formatting are structural, not prompted. A FastAPI layer (`main.py`) wraps the graph with request logging, input validation, and a health/teams debug endpoint; a Streamlit chat UI (`streamlit_app.py`) sits on top of the API, with an in-process fallback if the API isn't running. Every tool/model call runs with a 10s timeout and a bounded retry; every node is wrapped so an exception degrades to a clarification or fallback message instead of crashing the request.

3. Evaluation Results

Category	Passed	Total	Pass Rate
Factual Q&A	8	8	100.0%
Prediction Sanity	6	6	100.0%
Scope Guardrails	6	6	100.0%
Multi-Turn Coherence	3	5	60.0%
Overall	23	25	92.0%

25 cases ran against the live hardened graph (not hand-simulated): 8 factual Q&A, 6 prediction sanity checks, 6 scope-guardrail probes (off-topic + 2 prompt-injection attacks), and 5 multi-turn follow-ups. **Weakest category: Multi-Turn Coherence (60%)**. Two follow-ups failed: a team-only follow-up ("And what about against Geelong?") resolved to that team's season record instead of continuing the prior head-to-head, and a season-only follow-up ("What about in 2015?") had no AFL keyword for the router's guardrail to recognize and was refused as off-topic. Both share one root cause: `conversation_history` is now correctly threaded and stored per conversation (fixing the Day 4 gap), but the retrieval and router nodes don't yet read it back to resolve pronouns/ellipsis. **Fix:** add a lightweight context-resolution step before routing (reusing Day 3's `resolve_context()` pattern) that fills in an omitted team, season, or round from the most recent turn before the intent classifier and retrieval node run.

Prediction model vs. naive baseline (temporal holdout, most recent 20% of matches):

Model	Accuracy
Naive baseline (lower ladder position wins)	64.1%
Trained logistic-regression match-winner model	67.6%
Lift over baseline	+3.5 pp

Interpretation: the model beats the ladder-only baseline by 3.5 points, showing real (if modest) lift from the recent-form, rest-days, and win-streak features beyond ladder position alone. Prediction sanity checks confirm this qualitatively: a clear favorite (Sydney vs. Gold Coast) returns a high-confidence probability, while an even-ish matchup (Collingwood vs. Geelong) returns a probability much closer to 50/50.

4. Known Limitations

- **Data recency:** covers 2012-2018 only; no current-season fixtures or form.

- **Model accuracy ceiling:** ~68% on a temporal holdout — useful signal, not a reliable betting tool.
- **Multi-turn context:** history is stored but not yet used to resolve pronoun/ellipsis follow-ups (see §3).
- **Guardrail edge cases:** keyword-based off-topic/injection detection can miss novel phrasings a live LLM router would catch by reasoning about intent instead.

5. Monitoring & Maintenance Plan

Metric	Alert Threshold	Cadence
Response latency (p95)	> 10s	Real-time
Tool error rate	> 5%	Real-time
Off-topic leak rate	> 10%	Daily
Prediction accuracy	Drop > 5% vs. baseline	Weekly
Cost per conversation	> \$0.05	Daily
Token usage drift	> 20% change	Weekly

Weekly retraining loop:

- **Mon** — ingest new match results from the weekend.
- **Tue** — refresh the feature table (`features_v2.parquet`).
- **Wed** — re-evaluate prediction models on a fresh hold-out.
- **Thu** — if accuracy drops >5% vs. the running baseline, trigger retraining.
- **Fri** — deploy the updated model only if hold-out accuracy improves.

Re-evaluation cadence: guardrail tests weekly (25-case suite above, re-run against the current graph); prediction models weekly during season, using the loop above; full system review monthly (architecture, cost, and a manual re-read of a sample of logged conversations).

6. Recommended Next Steps

- Add the context-resolution fix above to close the multi-turn coherence gap.
- Continuous retraining on fresh results once a live data feed is available.
- Expand data sources beyond 2012-2018 and add current-season fixtures.
- Run a fairness review across teams/players before any public-facing deployment.
- Cost optimization: cache repeated factual/retrieval lookups; batch router calls where possible.